CHRISTINE YE SHU BLACKSHAW

Princeton University | Civil and Environmental Engineering | Princeton, NJ

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RESEARCH INTERESTS

Tropical cyclone hazards and risk; multi-hazard forecasting and uncertainty; climate change effects on extreme winds and coastal flooding; compound flood and extreme-heat hazards; urban and infrastructure resilience; climate and hazard modeling.

EDUCATION

PRINCETON UNIVERSITY, Princeton, NJ **2021-present**
Ph.D. Candidate, Civil and Environmental Engineering Anticipated 2027

- Advisor: Professor Ning Lin, Hurricane Hazards and Risk Analysis Group
- M.A. awarded May 2023 following completion of the qualifying examination

COLUMBIA UNIVERSITY SCHOOL OF ENGINEERING AND APPLIED SCIENCE, New York, NY **2017-2021**
B.S., Civil Engineering; Minor in Environmental Engineering

RESEARCH EXPERIENCE

PRINCETON UNIVERSITY, Princeton, NJ **2021-present**
Ph.D. Candidate / Researcher, Hurricane Hazards and Risk Analysis Group

- Lead research on tropical cyclone multi-hazard forecasting, climate-sensitive structural design winds, compound urban hazards, and synthetic hurricane track modeling.
- TC Compound Hazard Model Integration for Public Forecasting Website (lead): integrate official forecasts and hazard models to support real-time tropical cyclone multi-hazard assessment.
- Compound TC Flood and Extreme Heat Modeling in Urban Environments (lead): use atmospheric, urban-canopy, and flood modeling to examine interacting post-storm heat and inundation hazards.
- Structural Design Wind Estimations Under Future Climates (lead): estimate tropical cyclone design winds using synthetic storms, reanalysis, and CMIP6 climate-model ensembles.
- Wind Driven Synthetic Hurricane Track Modeling (lead): investigate synthetic tropical cyclone track modeling for wind-hazard applications.
- Contribute to research on tropical cyclone impacts on power systems with renewable integration.
- Past research includes paleo-hurricane reconstruction from hydrodynamic modeling in The Bahamas and historical reconstruction of urban storm and meteorological events in Camden, New Jersey.

COLUMBIA UNIVERSITY, New York, NY **2020-2021**
Student Researcher, Department of Civil Engineering and Engineering Mechanics

- Constructed a three-dimensional leaf-area-density model for Manhattan using LiDAR, tree-census data, and USDA street-tree growth projections; reviewed remote-sensing databases and resources.
- Developed a subway flood-propagation and damage model for research on hurricane flood protection in New York City; modeled storms and surface flooding in Python and visualized surface and subway inundation in ArcGIS.

TEACHING AND MENTORING EXPERIENCE

PRINCETON UNIVERSITY **Spring 2024**
Assistant in Instruction, CEE 460: Risk Analysis

PRINCETON UNIVERSITY **Spring 2023, 2024**
Assistant in Instruction, CEE 306: Hydrology: Water and Climate

PRINCETON UNIVERSITY **Summers 2022-2025**
Undergraduate and High School Research Mentor

POLYGENCE **2023-present**
Research Mentor

AWARDS AND HONORS

SEAS Award for Excellence **2024**

NSF Graduate Research Fellowship **2022-2026**

Tullio J. Borri '51 Award in Civil Engineering **2021**

Columbia University Dean's List **2018-2021**

PUBLICATIONS AND MANUSCRIPTS

Journal articles, manuscripts, and reports

1. **Blackshaw, C. Y.**, & Lin, N. (2026). Climate Sensitivity of Tropical Cyclone Design Winds in the United States. Earth's Future. Under review.
2. **Blackshaw, C. Y.**, Spaulding, A. C., & Lin, N. (2026). Real-Time Tropical Cyclone Multi-Hazard Forecasting. Geophysical Research Letters. In preparation.
3. Xu, L., Lin, N., Perera, A. D., Spaulding, A. C., Wang, Z., Oppenheimer, M., **Blackshaw, C. Y.**, & Poor, H. V. (2026). Climate change exacerbates disparities of energy resilience in New York City. Nature Communications.
4. **Blackshaw, C. Y.**, Lin, N., Begmohammadi, A., & Bou-Zeid, E. (2025). Investigating the interaction of tropical cyclone-heatwave compound hazards in urban environments. Urban Climate, 64, 102667.
5. Miura, Y., **Blackshaw, C. Y.**, Zhang, M. S., Mandli, K. T., & Deodatis, G. (2025). Coastal storm-induced flooding risk of the New York City subway amid climate change. Transportation Research Part D: Transport and Environment, 149, 104974.
6. Begmohammadi, A., Lin, N., Xi, D., & **Blackshaw, C.** (2025). Assessing future coastal flood hazards from tropical cyclones in the Northeastern United States. Earth's Future, 13(11), e2025EF006063.
7. Begmohammadi, A., **Blackshaw, C. Y.**, Lin, N., Gori, A., Wallace, E., Emanuel, K., & Donnelly, J. P. (2024). Integrating Climatological-Hydrodynamic Modeling and Paleohurricane Records to Assess Storm Surge Risk. Journal of Geophysical Research: Oceans, 129(1), e2023JC020354.
8. Princeton University. (2022). Expanding Climate Services to Respond to Adaptation Challenges: Insights from Managed Retreat Practice. Policy Workshop Report. Princeton School of Public & International Affairs.

PRESENTATIONS

1. **Blackshaw, C.**, & Lin, N. (2025). Designing for Tropical Cyclone Wind Loads under Non-Stationary Climates. AGU Fall Meeting 2025.
2. **Blackshaw, C.**, Begmohammadi, A., Lin, N., & Bou-Zeid, E. (2024). Investigating the interaction of tropical cyclone-heatwave compound hazards in urban environments. AGU Fall Meeting 2024.
3. **Blackshaw, C.** (2024). Paleo-hurricane Data for Long-term Hurricane Risk Assessment. Invited panel presentation, WTW Natural Catastrophe Program.
4. **Blackshaw, C.**, Begmohammadi, A., Lin, N., & Bou-Zeid, E. (2024). Investigating the Interaction of Tropical Cyclone-Heatwave Compound Hazards in Urban Canopy Models. Columbia University Extreme Heat Workshop.
5. **Blackshaw, C.**, Begmohammadi, A., & Lin, N. (2023). Integrating Climatological-Hydrodynamic Modeling and Paleohurricane Records to Assess Storm Surge Risk. AGU Fall Meeting 2023.
6. **Blackshaw, C.**, Begmohammadi, A., & Lin, N. (2023). Integrating ADCIRC modeling and paleohurricane records to assess long-term storm surge risk. ADCIRC Users Group Meeting 2023.

ACADEMIC AND COMMUNITY ENGAGEMENT

PRINCETON ENVIRONMENT AND CLIMATE SCHOLAR **2024-2026**
Research Project Lead; Faculty Seminar Coordinator

- Research project lead (2025-2026): evaluate cooling and energy demands across Princeton campus and situate campus actions within the broader energy-policy landscape.
- Faculty seminar coordinator (2024-2025): organize interdisciplinary climate seminars with faculty and students.

PRINCETON CIVIL AND ENVIRONMENTAL ENGINEERING GRAD COUNCIL **2023-present**
Member

PROFESSIONAL DEVELOPMENT AND WORKSHOPS

- SPI 591G: Climate Adaptation and the Coasts: Managed Transition (Fall 2021)
- LabTales: A Science Storytelling Training Workshop (Summer 2022)
- Investing in Futures Graduate Cohort (Fall 2022)
- SPI 586E: Topics in STEP - Natural Hazards and Disaster Policy (Fall 2024)
- AAAS CASE Workshop (Spring 2025)
- Munich Re Externship (Spring 2026)

TECHNICAL EXPERTISE

Programming & computing: Python, R, MATLAB, AWS, HPC workflows, HTML
Geospatial & visualization: GIS, ArcGIS, ParaView
Hazard & climate models: PepC, ADCIRC, SFINCS, HEC-RAS, WRF, PUCM, tropical cyclone rainfall and wind modeling, global climate models

ACTIVITIES AND INTERESTS

Company Member, The Abdul Latif Collective (TALC) **2021-present**

Princeton Dance Festival; Princeton University Ballet **2021-present**

Columbia/Barnard Dance Department Performances; Columbia Ballet Collaborative; Columbia Repertory Ballet; Columbia University Ballet Ensemble **2018-2021**